Inmon vs Kimball Data Warehouse Approaches

Inmon Approach

The Inmon approach follows a **top-down methodology**. It starts by building a centralized Enterprise Data Warehouse (EDW) using a normalized model (3NF). Data marts are created later from the EDW.

Key characteristics:

- Top-down approach
- Normalized (3NF) data model
- Centralized enterprise data warehouse
- High data consistency and integration

Advantages

- High data consistency
- No redundancy
- Easy to maintain long-term

Disadvantages

- Slow to build
- Complex
- Expensive

Kimball Approach

The Kimball approach follows a **bottom-up methodology**. It starts by building dimensional data marts using star schema. These data marts are later integrated using conformed dimensions to form an enterprise data warehouse.

Key characteristics:

- Bottom-up approach
- Dimensional modeling (star schema, snowflake)
- Fast implementation
- Focus on business processes

Advantages:

- Faster implementation
- Easy to understand
- Good for reporting

Disadvantages:

- Data redundancy
- Hard to maintain consistency
- No single source of truth

Comparison

Inmon vs Kimball:

Feature	Inmon	Kimball
Approach	Top-down	Bottom-up
Model	Normalized (3NF)	Dimensional (Star Schema)
Start Point	Enterprise DW	Data Marts
Speed	Slow	Fast
Complexity	High	Lower
Use Case	Large enterprises	Reporting and analytics

Conclusion

Both produce **EDW**, but:

- Inmon → **physical central EDW first**
- Kimball → **virtual/logical EDW after integration**

- Inmon: EDW (3NF) → Data Marts (Dimensional)
- Kimball: Data Marts (Dimensional) → Integrated → EDW (Dimensional)

Full Load vs Incremental Load in Data Warehouse

1. Full Load

What is it? - Load **ALL data every time** from source → DW

Example:

Source table = 1,000,000 rows
Full Load → load all 1,000,000 rows again

How it works:

- Delete old data (or overwrite)
- Insert everything again

Advantages:

- Simple to implement
- No need to track changes
- Always consistent (fresh data)

Disadvantages:

- Very slow for large data
- High resource usage
- Not scalable

When to use:

- Small datasets
- First time loading (Initial Load)
- Testing

2. Incremental Load

What is it? - Load **ONLY new or changed data**

Example:

Yesterday: 1,000,000 rows
Today: +5,000 new rows

Incremental Load → load only **5,000 rows**

How it works:

Detect changes using:

- Timestamp (last updated column)
- ID (max ID)
- CDC (Change Data Capture)

Advantages:

- Very fast
- Efficient
- Scalable for big data

Disadvantages:

- More complex logic
- Needs tracking changes
- Risk of missing data if logic is wrong

When to use:

- Large datasets
- Daily / hourly ETL jobs
- Production systems

Conclusion

Feature	Full Load	Incremental Load
Data Loaded	All data	Only new/changed
Speed	Slow	Fast
Complexity	Simple	Complex
Data Volume	High	Low
Use Case	Initial load	Daily updates

- **Real-World Flow in DW**

Usually BOTH are used together:

1. First time → **Full Load**
2. After that → **Incremental Load**